BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Integrated Development Environments

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does an interpreter normally do?

2. Q = (A AND B) OR C. A=1, B=0, C=1. What is Q?

A) 0 B) 2
C) 1 D) 10

3. "A count-controlled loop is only suitable when the number of repetitions can never be known." Correct this. Explain why.

4. A program keeps asking for a password until its stopping condition is met. "A condition-controlled loop always runs ten times." Correct this. What controls when it stops?

Today's objectives

Explain the core ideas of integrated development environments using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Editor

Provides tools for entering and organising source code.

Syntax highlighting and completion

Make code structure and possible mistakes easier to notice.

Breakpoints and debugging

Pause execution to inspect variables and follow faulty logic.

Runtime/output tools

Allow programmers to run/test code and view results from the environment.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Editor	Provides tools for entering and organising source code.
Syntax highlighting and completion	Make code structure and possible mistakes easier to notice.
Breakpoints and debugging	Pause execution to inspect variables and follow faulty logic.
Runtime/output tools	Allow programmers to run/test code and view results from the environment.

I Do - One worked example

A program produces 9 instead of expected 10. Explain how a debugger could locate the error.

1. Set a breakpoint before the suspect calculation.
2. Run to the breakpoint.
3. Inspect current variable values and step line by line.
4. Find first unexpected update, fix it and rerun tests.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A pupil has misspelled a variable. Identify two IDE features that may help.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the faulty bits-to-bytes calculation and check its output.

```
bits = 40
bytes = bits * 8
print(bytes)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Name two IDE features. [2 marks]

Q3. Explain why a breakpoint helps a developer. [3 marks]

Q4. State why a program should be tested after fixing an IDE-reported syntax error. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A breakpoint changes all incorrect variables automatically. ☐ T ☐ F
2. Provides tools for entering and organising source code. ☐ T ☐ F
3. Make code structure and possible mistakes easier to notice. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 An interpreter translates and runs code as it goes, often one statement at a time. This gives quick feedback during testing. ☐

Q2 C - 1 ☐

Q3 When the number of repetitions is known or determined by a fixed count. ☐

Q4 A condition-controlled loop repeats until a condition tells it to stop. It need not run ten times. Choose the loop to suit the task and trace its steps. ☐

Code First - Model

```
bits = 40
bytes = bits DIV 8
print(bytes)
```

Output: 5

We Do - Check the model

1. Test after correction. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Code editor, syntax highlighting, autocomplete, debugging, breakpoints, error diagnostics or run console. ☐

Q3 Pauses execution at a selected point, allowing variables/program state to be inspected. ☐

Q4 Syntax correction only ensures code can be parsed; logical behaviour still needs checking. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: